

St. Francis Institute of Technology, Mumbai-400 103
Department of Information Technology

A.Y. 2024-2025

Class: BE Honours, Semester: VII

Subject: **AI-ML in Healthcare Lab**

Experiment – 2: To implement Data Visualization/Data Exploratory Analysis using Matplotlib and Seaborn on Healthcare dataset.

1. Aim: To implement Data Visualization using Matplotlib and Seaborn on healthcare dataset.

2. Objectives: After study of this experiment, the student will be able to

- Understand Matplotlib Functions
- Understand Seaborn Functions

3. Outcomes: After study of this experiment, the student will be able to

- Understand data visualization /Exploratory Data Analysis using Matplotlib and Seaborn.

4. Prerequisite: Fundamentals of Python Programming and Database Management System.

5. Requirements: Python Installation, Personal Computer, Windows operating system, Internet Connection, Microsoft Word.

6. Pre-Experiment Exercise:

Brief Theory:

Basic Concepts of Matplotlib and Seaborn.

7. Laboratory Exercise

A. Procedure:

```
import numpy as np
a=np.array([[1,2,3],[4,5,6],[7,8,9]])
a

"""**matplotlib**

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations
in Python. Matplotlib makes easy things easy and hard things possible.

1. Create publication quality plots.
2. Make interactive figures that can zoom, pan, update.
3. Customize visual style and layout.
4. Export to many file formats .
5. Embed in JupyterLab and Graphical User Interfaces.
6. Use a rich array of third-party packages built on Matplotlib.

**matplotlib.pyplot**

matplotlib.pyplot is a collection of functions that make matplotlib work like MATLAB. Each pyplot
function makes some change to a figure: e.g., creates a figure, creates a plotting area in a
figure, plots some lines in a plotting area, decorates the plot with labels, etc.

**numpy.pi**

While both numpy.pi and math.pi provides us with the same value, the advantage of numpy.pi is that
it helps us by avoiding the dependency of obtaining the constant value of pi from a different
library.

**np.sin**

This mathematical function helps user to calculate trigonometric sine for all x (being the array
elements).
"""

# Sinusoidal plot
import numpy as np
import matplotlib.pyplot as plt
x=np.arange(0.0,2.0,0.01)
y=np.sin(2*np.pi*x)
plt.plot(x,y)

#Barplot
import numpy as np
import matplotlib.pyplot as plt
fig=plt.figure()
x=[1,2,3,4,5,6,7]
y=[23,43,65,32,87,45,90]
plt.bar(x,y,color='orange')
plt.title('Average Score of Students')
plt.xlabel("Students")
plt.ylabel("Avg. Score")
```

```
fig.savefig("testing.jpg")
```

```
"""**Error Bars**
```

An error bar is a line through a point on a graph, parallel to one of the axes, which represents the uncertainty or variation of the corresponding coordinate of the point.

Error bars can communicate the following information about your data:

1. How spread the data are around the mean value (small SD bar = low spread, data are clumped around the mean; larger SD bar = larger spread, data are more variable from the mean).
2. The reliability of the mean value as a representative number for the data set. In other words, how accurately the mean value represents the data (small SD bar = more reliable, larger SD bar = less reliable). It's important to note that just because you have a larger SD, it does not indicate your data is not valid.
3. The likelihood of there being a significant difference between data sets.

```
**Significant Difference**
```

A "significant difference" means that the results that are seen are most likely not due to chance or sampling error. In any experiment or observation that involves sampling from a population, there is always the possibility that an observed effect would have occurred due to sampling error alone. But if result is "significant," then the investigator may conclude that the observed effect actually reflects the characteristics of the population rather than just sampling error or chance.

```
#single barplots with error bars
```

```
N=5
men=(20,35,30,36,27)
women=(25,32,34,20,25)
mstd=(2,2,2,2,2)
i=np.arange(N)
p1=plt.bar(i,men,width=0.35,yerr=mstd)
p2=plt.bar(i,women,width=0.35,bottom=men,yerr=mstd)
plt.xticks(i,('G1','G2','G3','G4','G5'))
plt.yticks(np.arange(0,90,10))
plt.legend((p1[0],p2[0]),('Men','Women'))
```

```
"""**Histogram**
```

A histogram is a graphical representation of a grouped frequency distribution with continuous classes.

A histogram is a diagram involving rectangles whose area is proportional to the frequency of a variable and width is equal to the class interval.

The histogram graph is used under certain conditions. They are:

1. The data should be numerical.
2. A histogram is used to check the shape of the data distribution.
3. Used to check whether the process changes from one period to another.
4. Used to determine whether the output is different when it involves two or more processes.
5. Used to analyse whether the given process meets the customer requirements.

Histograms and bar charts are different. In the bar chart, each column represents the group which is defined by a categorical variable, whereas in the histogram each column is defined by the continuous and quantitative variable.

```
**Difference Between Histogram and Bar Graph**
```

1. It is a two-dimensional figure
2. The frequency is shown by the area of each rectangle
3. It shows rectangles touching each other

```
**Bar Graph**
```

1. It is a one-dimensional figure
2. The height shows the frequency and the width has no significance.
3. It consists of rectangles separated from each other with equal spaces.

```
"""
```

```
#Histogram
```

```
m=100
std=15
x=m+std*np.random.randn(480)
y=50
plt.hist(x,y,color='blue',density=False)
plt.grid(color='gray',linestyle='--',linewidth=2,axis='x')
plt.grid(color='gray',linestyle='--',linewidth=2,axis='y')
```

```
#Pie Chart
```

```
labels='cricket','Hockey','Tennis','Football'
v=[40,20,10,30]
e=(0,0,0.1,0)
plt.pie(v,explode=e,labels=labels,autopct='%1.3f%%',shadow=True,startangle=0)
```

```
"""numpy.linspace(start, stop, num)
```

Return evenly spaced numbers over a specified interval.

Returns num evenly spaced samples, calculated over the interval [start, stop].

The endpoint of the interval can optionally be excluded.

```
"""
```

```
x=np.linspace(0,10,1000)
plt.plot(x,np.sin(x))
plt.plot(x,np.sin(x-1),color='g')
plt.plot(x,np.sin(x-2),color='0.25')
```

```

plt.plot(x,np.sin(x-3),color='red')

x=np.linspace(0,10,1000)
plt.plot(x,np.sin(x))
plt.plot(x,np.sin(x-1),linestyle='solid')
plt.plot(x,np.sin(x-2),linestyle='dashed')
plt.plot(x,np.sin(x-3),linestyle='dashdot')
plt.plot(x,np.sin(x-4),linestyle='dotted')

#plt.plot(x,np.sin(x))
plt.plot(x,np.sin(x-1),linestyle='-')
plt.plot(x,np.sin(x-2),linestyle='--')
plt.plot(x,np.sin(x-3),linestyle='-.')
plt.plot(x,np.sin(x-4),linestyle=':')

#Scatterplot
x=np.linspace(0,10,50)
y=np.sin(x)
plt.scatter(x,y)

x=np.linspace(0,10,50)
plt.plot(x,np.sin(x),'-o')

#Seaborn
import seaborn as sns
data=sns.load_dataset("iris")
sns.lineplot(x="sepal_length", y="sepal_width",data=data)

sns.scatterplot(x="sepal_length", y="sepal_width",data=data)

sns.stripplot(x="species", y="sepal_length",data=data)

sns.violinplot(x="species", y="sepal_length",data=data)

sns.swarmplot(x="species", y="sepal_length",data=data)

```

Paste Screenshots of above commands and apply these commands on Insurance dataset.

8. Post-Experiments Exercise

A. Extended Theory: (Soft Copy)

What is difference between table and graph? (Ref Link 1)

B. Questions:

What is the difference between Matplotlib and Seaborn? (Refer Link 2)

C. Conclusion:

Write the significance of the topic studied in the experiment.

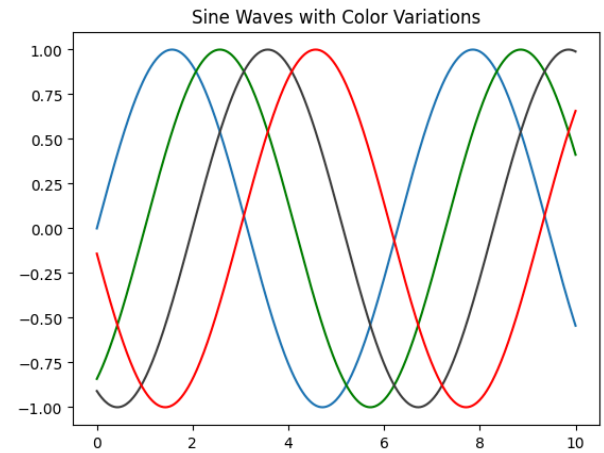
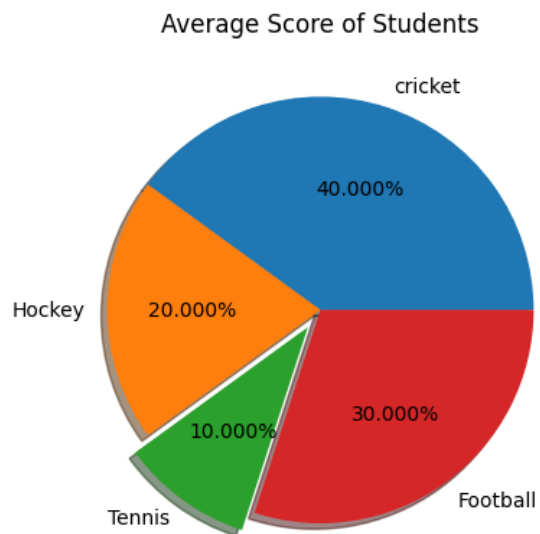
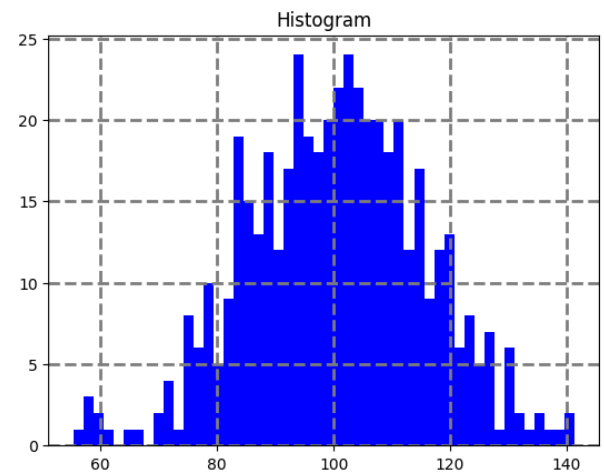
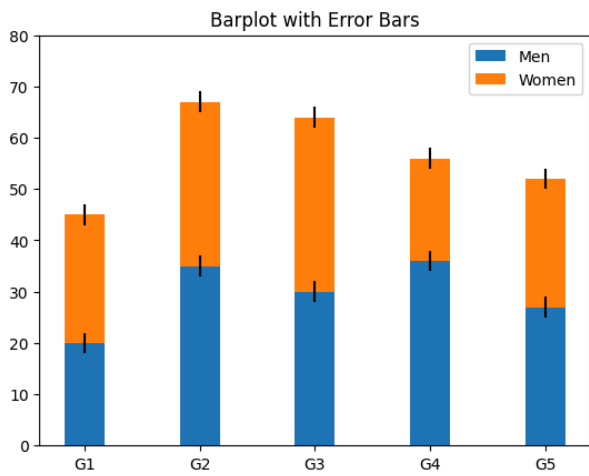
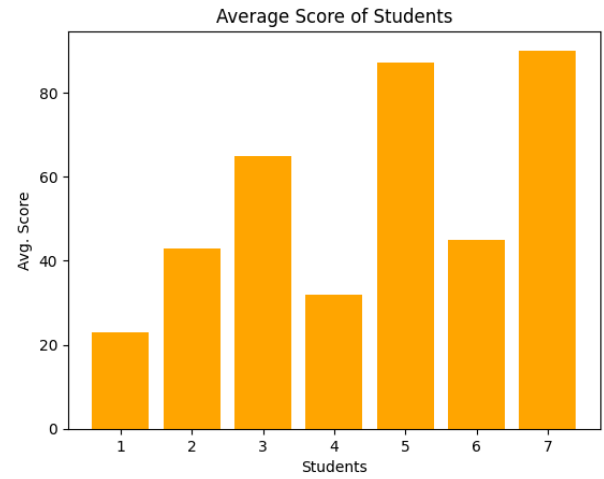
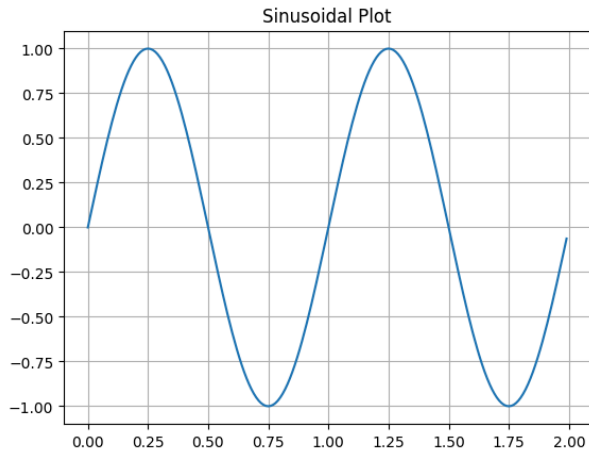
D. References:

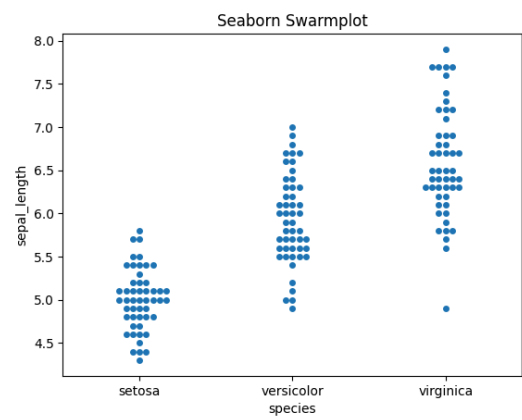
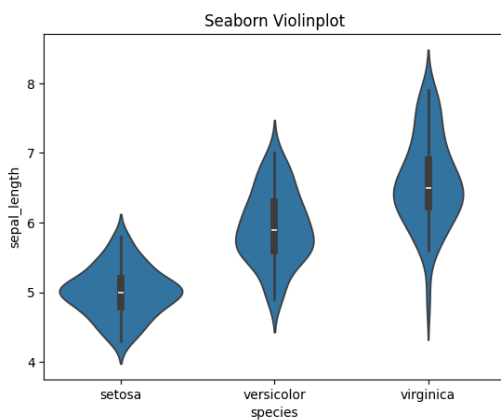
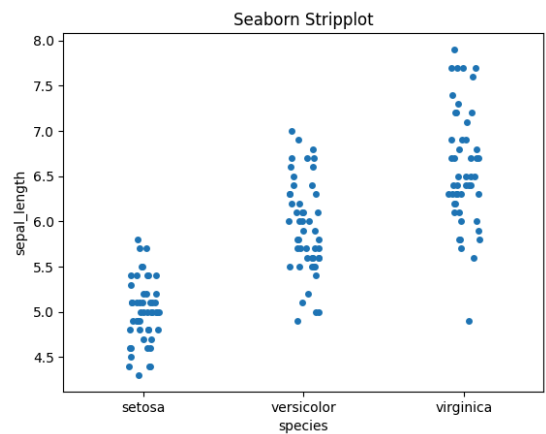
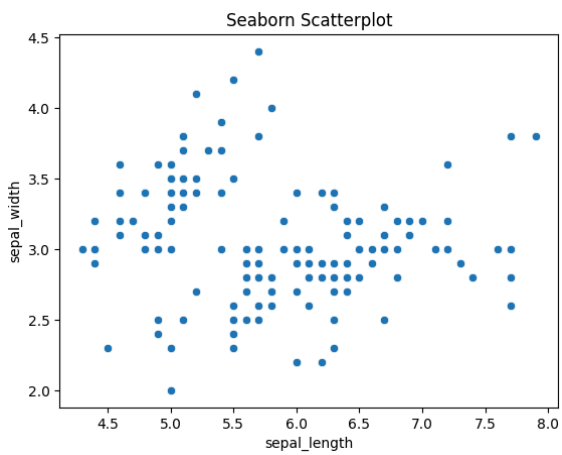
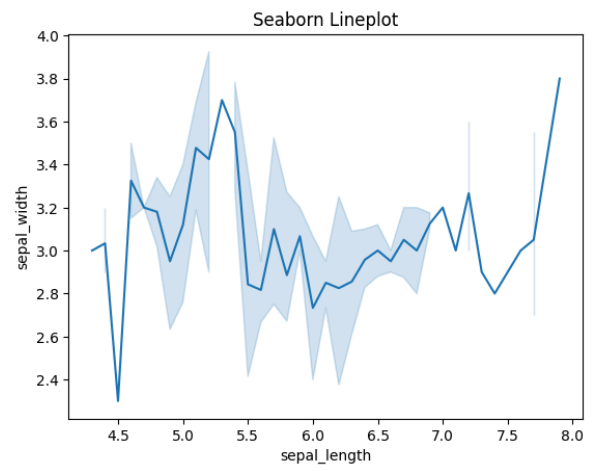
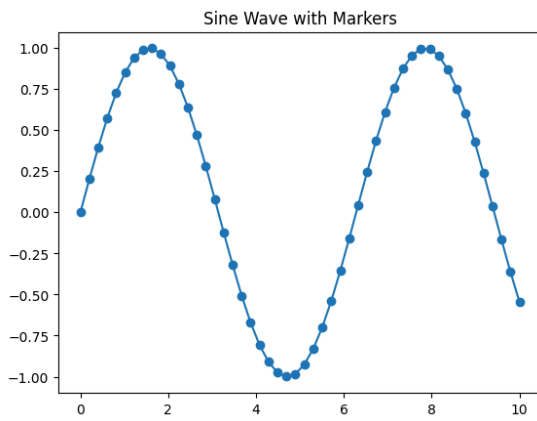
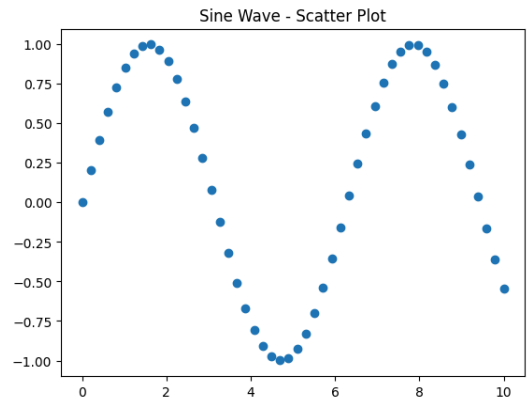
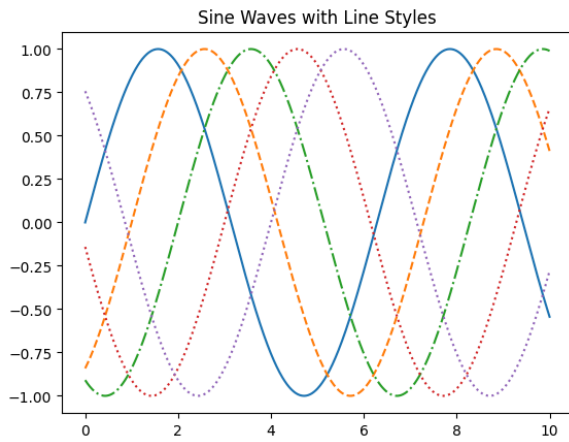
1. <https://www.wallstreetmojo.com/graphs-vs-charts/>
2. <https://sdsclub.com/seaborn-vs-matplotlib-python-visualization-tools-battle/>
3. <https://matplotlib.org/stable/tutorials/index>
4. <https://seaborn.pydata.org/>

Laboratory Exercise

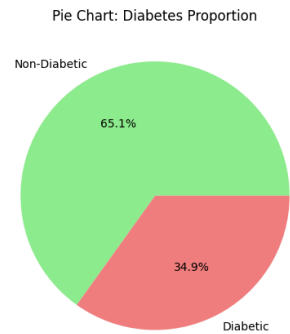
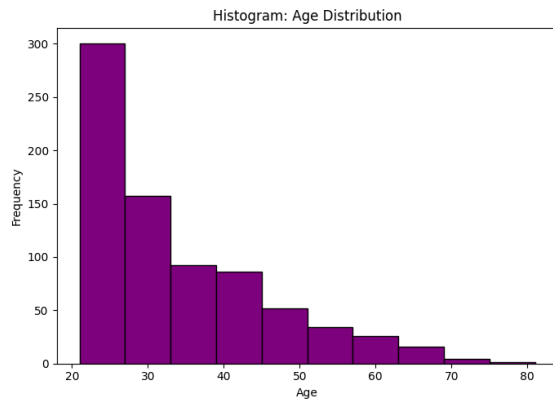
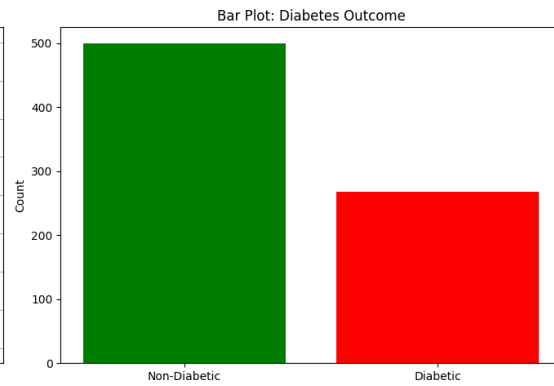
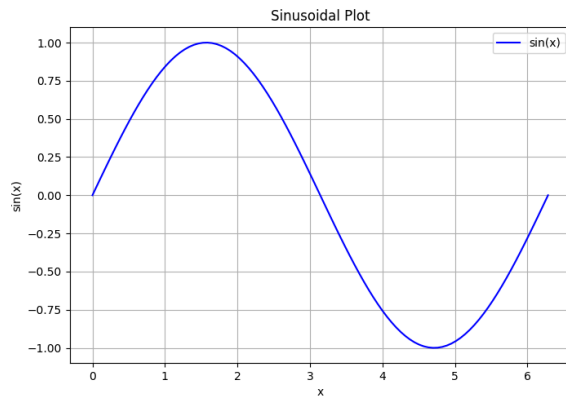
A. Procedure:

CELL 1 output:





OUTPUT: CELL 2(On insurance dataset)



8. Post-Experiments Exercise

A. Extended Theory: (Soft Copy)

What is difference between table and graph?

Table	Graph
A table organizes data systematically in rows and columns for easy reading and reference.	A graph represents data visually using lines, bars, or points to show relationships and trends.
Used to display exact numerical data for detailed comparison and analysis.	Used to visualize patterns, trends, or comparisons quickly.
Represents data in textual and numerical form.	Represents data pictorially or graphically.
Requires analytical reading to interpret data.	Easier to interpret visually at a glance.
Provides precise and specific data values.	Gives an approximate visual representation rather than exact numbers.
When accuracy and detail are more important than visualization.	When quick comparison or pattern identification is needed.
Frequency tables, financial statements, data matrices.	Bar charts, line graphs, pie charts, scatter plots.